

COL1000: Introduction to Programming

Lecture 1

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Department of CSE, IIT Delhi**

Course Administrivia

Course Evaluation

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1. Lab exams: $15\% + 15\% = 30\%$
2. Quiz (on 31st January): 5%
3. Minor exam: 25%
4. Major exam: 30%
5. Attendance: 5% (lectures) + 5% (labs)

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Link: <https://rollcall.iitd.ac.in/>

Course codes:

COL1000-416, COL1000-418, COL1000-111

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Course webpage

<https://web.iitd.ac.in/~keerti/Courses/COL1000-2026.htm>

Reference Books

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1. Introduction to Computation and Programming Using Python by John V. Guttag.
2. Think Python by Allen B. Downey.

General Course Advice

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4. Please do not cheat!

What you will learn in this course ...

**This course is not about learning Python
(although you will learn Python on the way)**

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1. How is **data stored** and **processed** in a computer?
2. How to write **clean, readable, error-free** code?
3. How to analyse the correctness and **efficiency** of your code?

An Informal Introduction to Computer Programming

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What is a computer?

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An electronic *programmable* device that can:



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- can accept input;
- process the data (**arithmetic and logical operations**);
- can **store data and results** for future use;
- produce output.

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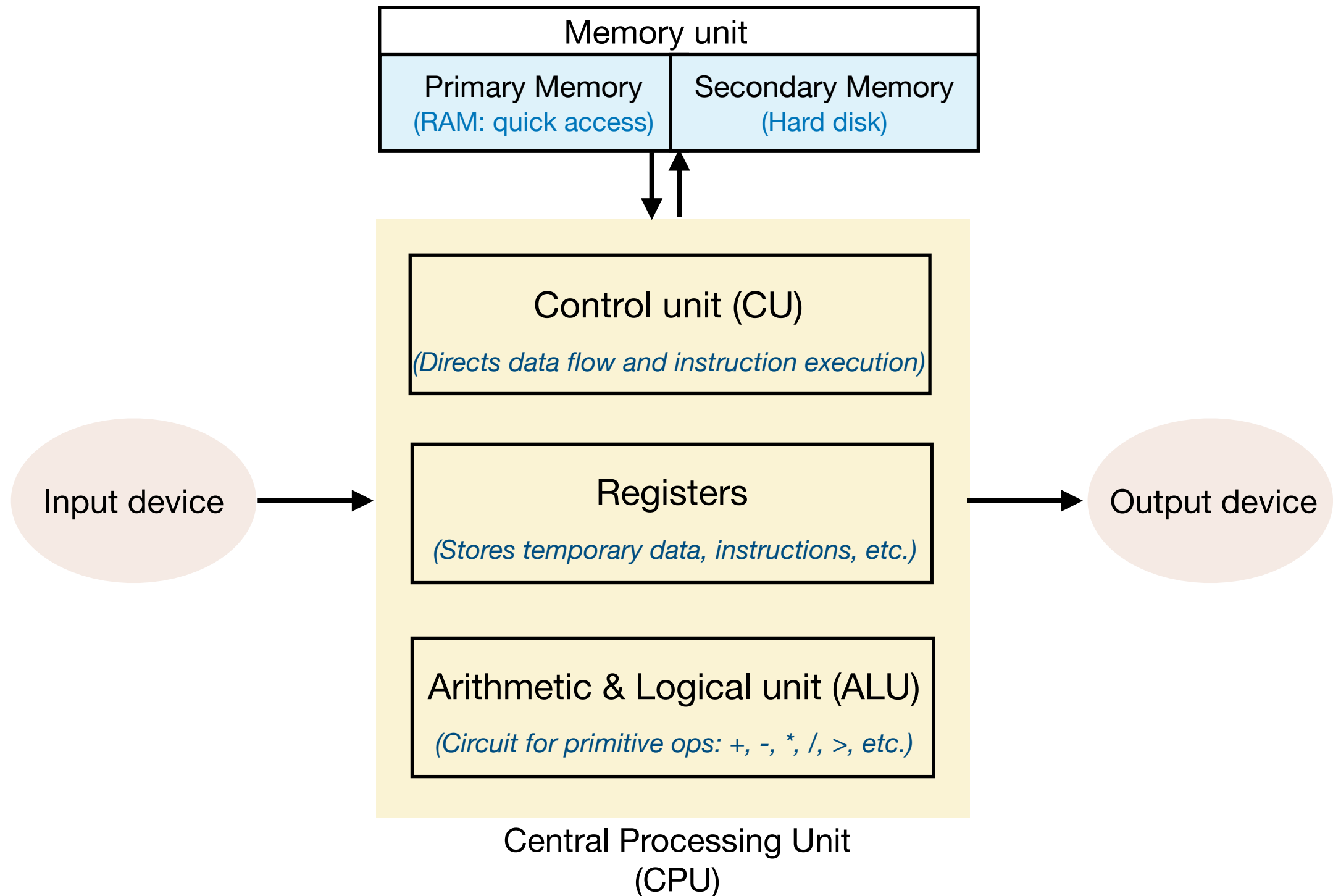
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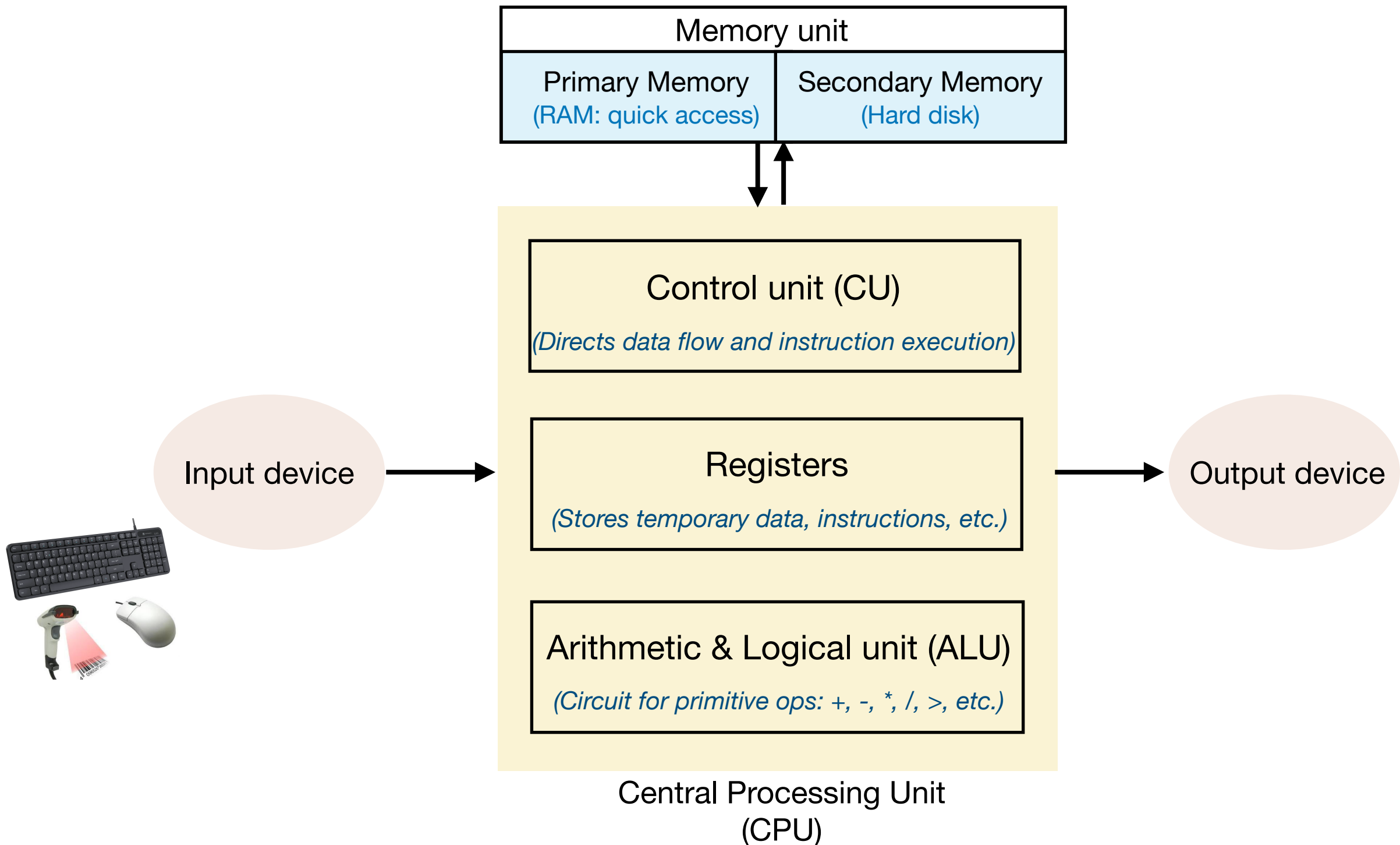
Computers can do **lots** of **fast** calculations, have **lots** of storage,
but need **precise** instructions

Main components of Computer Hardware

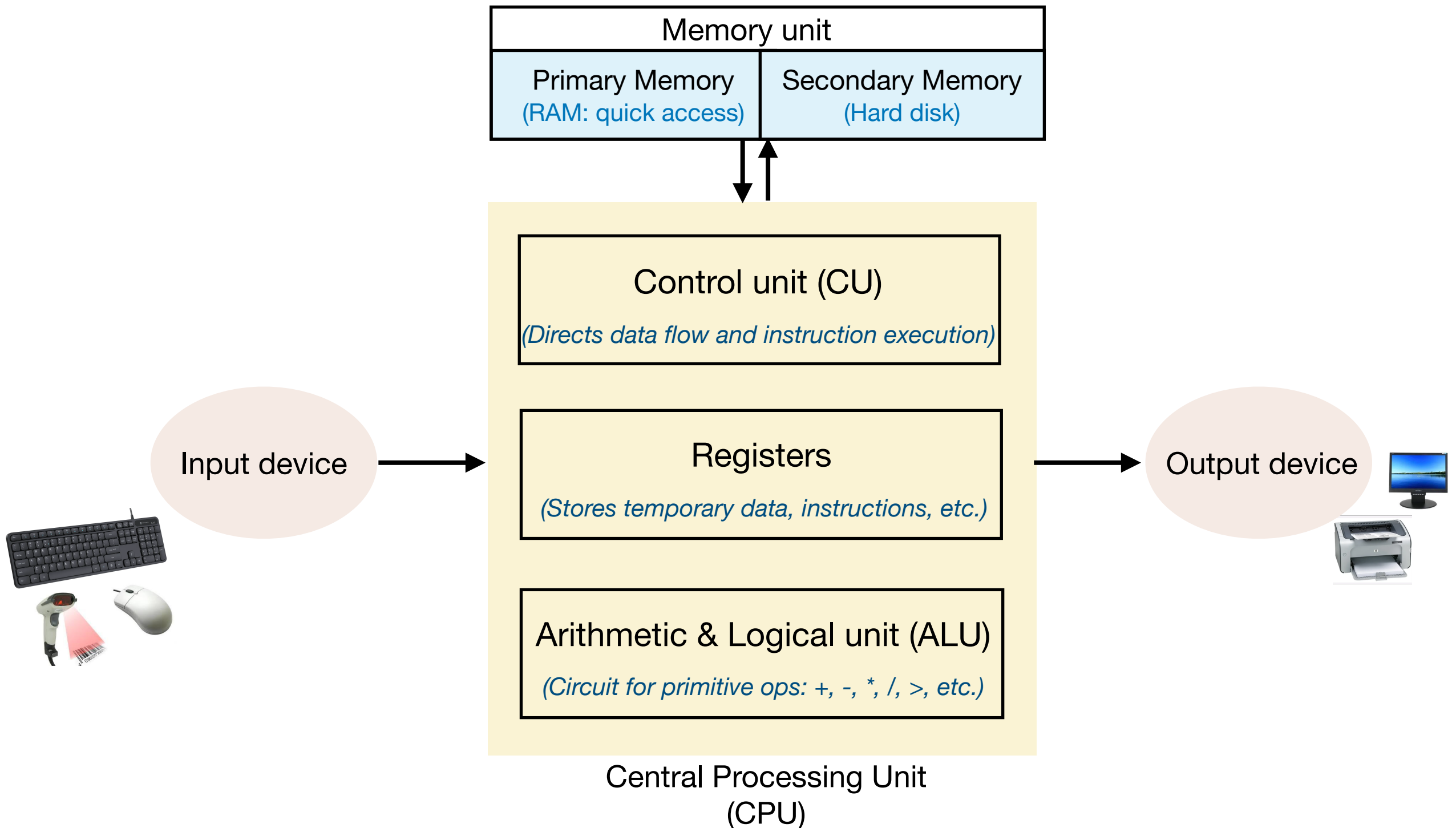
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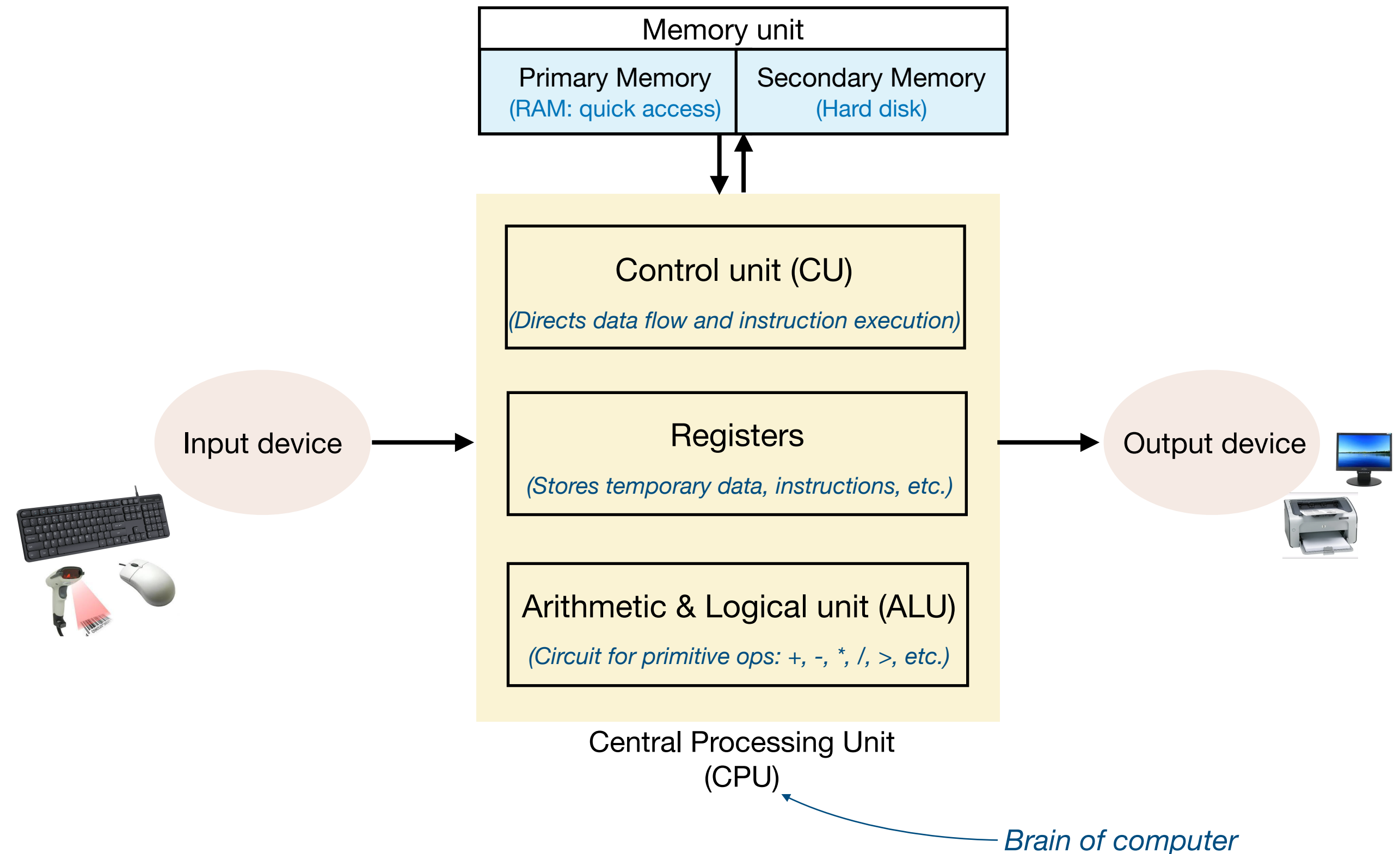
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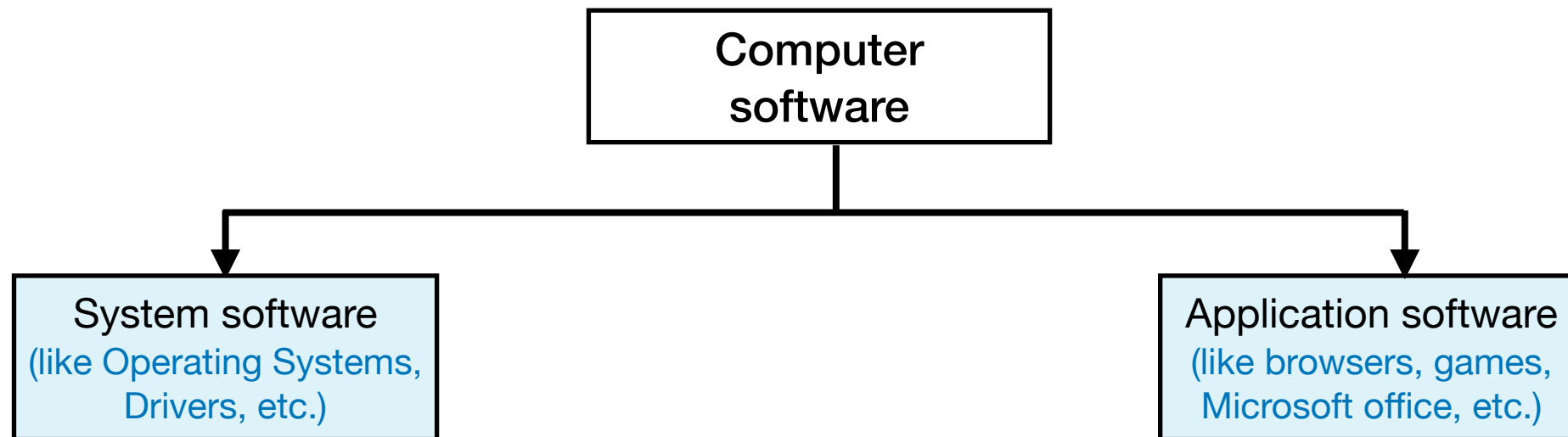


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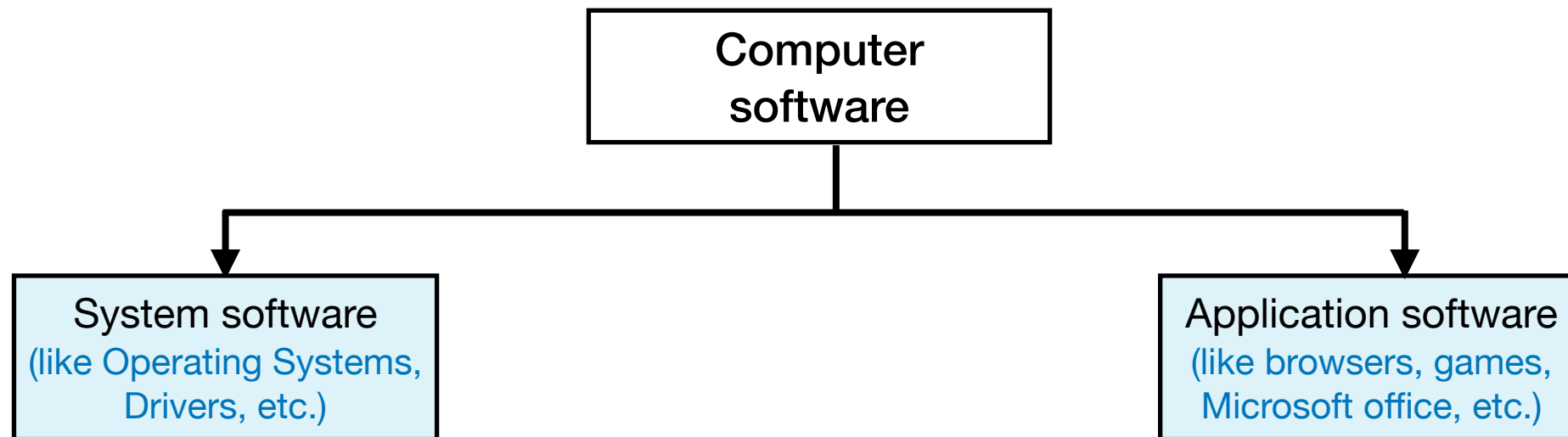


Main components of Computer Software

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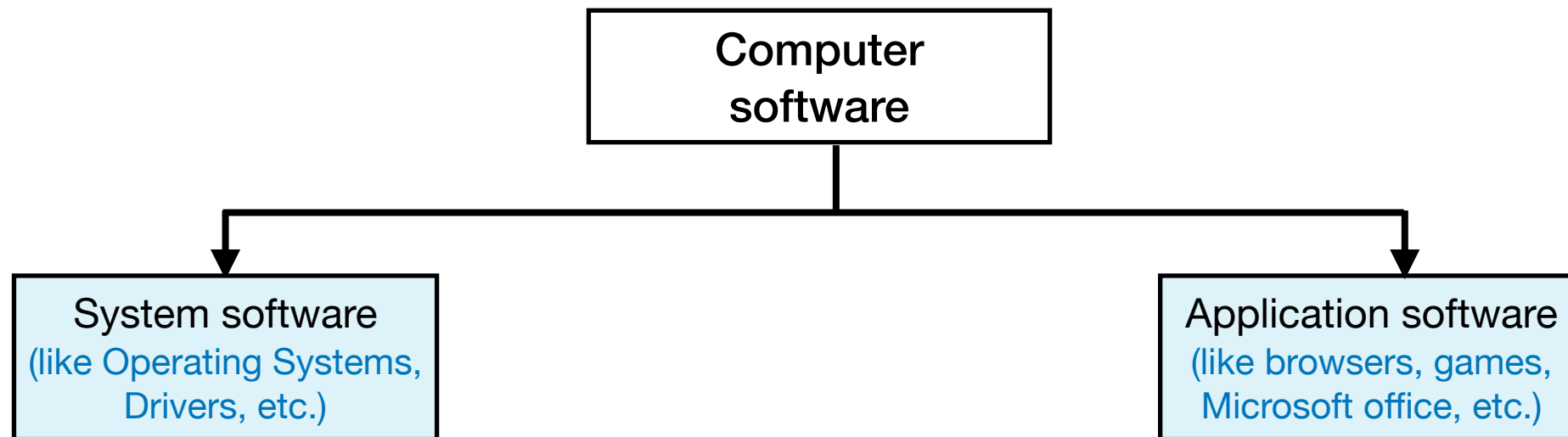


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Operating System

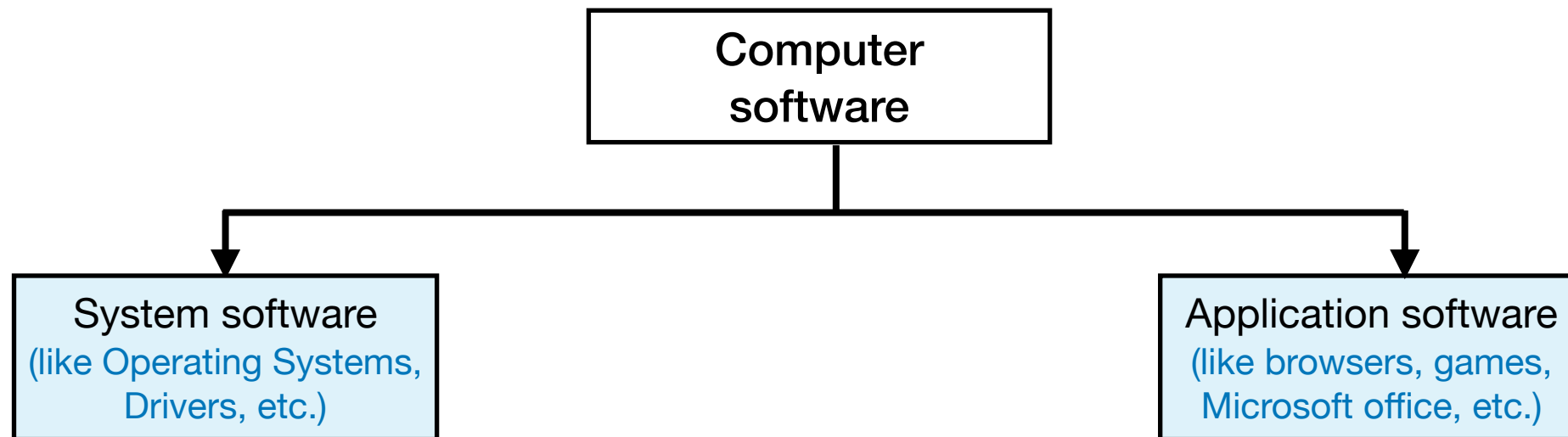
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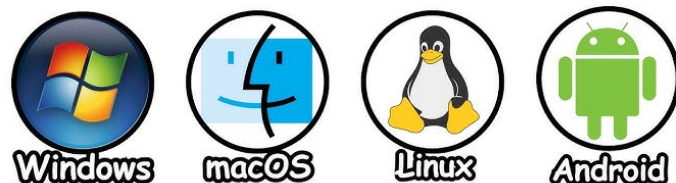
- **Interface** between computer's **hardware** and **user/applications**.
- Manages everything from **opening/running applications**, handling files, controlling mouse/keyboard.

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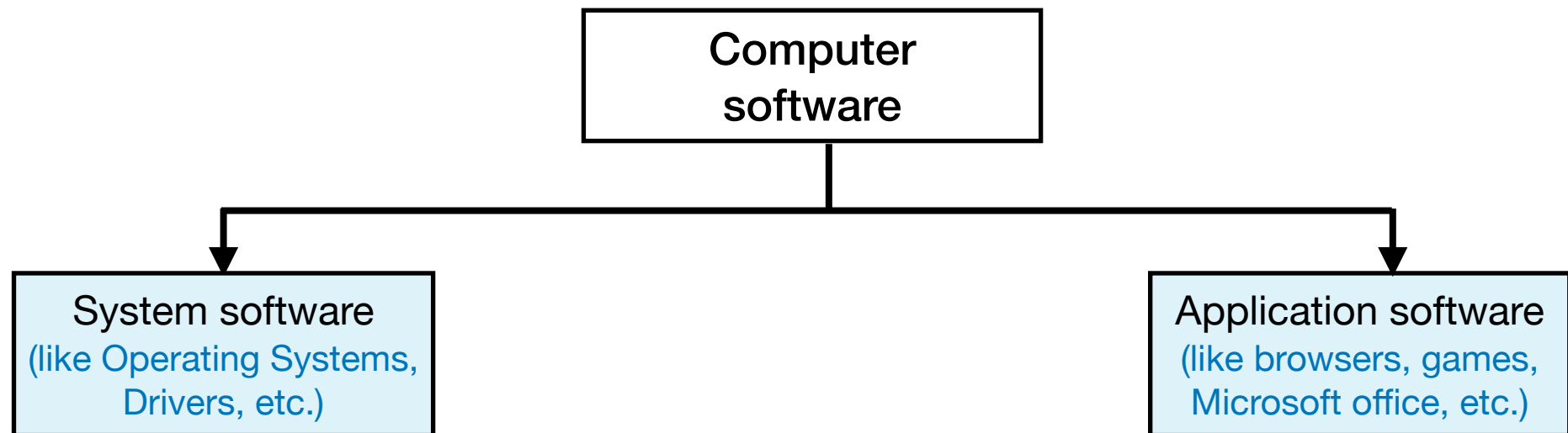


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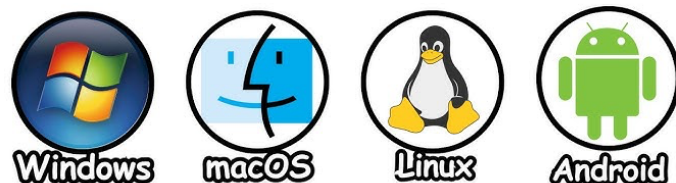
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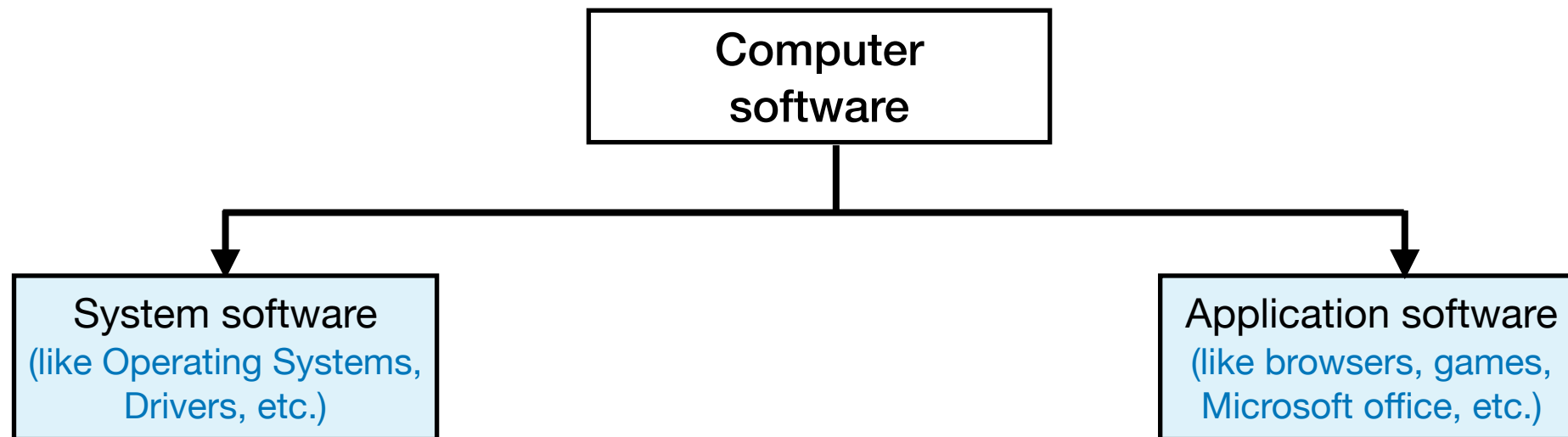
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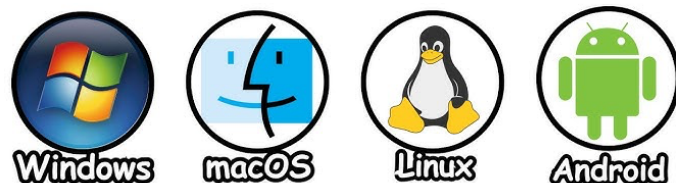


All applications run on **top** of the operating system.

All **applications** are **written** in **programming language**, and **python** is one of them.

Operating System

- **Interface** between computer's **hardware** and **user/applications**.
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Why use Programming language?

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- The CPU understands only machine language (binary code).

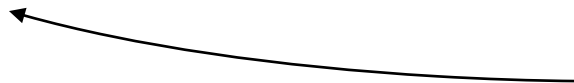
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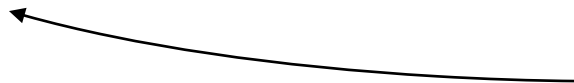
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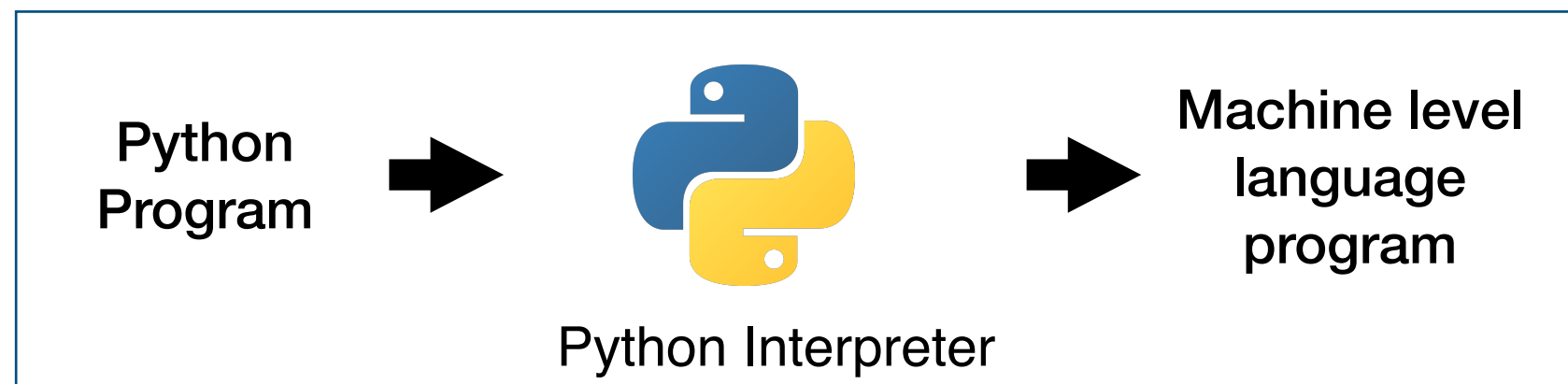
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- Programming language — closer to human language.
- **Compiler/Interpreter** converts “program language code” to “machine-level language code”.



An Informal Introduction to Computer Programming

What is a Program?

A program is a “sequence of instructions” that a computer can execute to perform a task.

An algorithm (aka. program) from high school:

Addition of Large Numbers

$$\begin{array}{r} 73407 \\ + 61725 \\ \hline \end{array}$$

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An algorithm (aka. program) from high school:

Addition of Large Numbers

$$\begin{array}{r} 73407 \\ + 61725 \\ \hline 2 \end{array}$$

The diagram illustrates the addition of two large numbers, 73407 and 61725, using a high school algorithm. The numbers are aligned by their least significant digits (units). A red '1' is positioned above the units column, indicating a carry-in from a previous step. The units column (7 + 5) is highlighted with an orange background. A horizontal line is drawn under the numbers, and the result '2' is shown below the line, representing the sum of the units column (7 + 5 = 12, with 2 being the units digit and 1 being the carry).

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 1 \\ 7 \ 3 \ 4 \ 0 \ 7 \\ + 6 \ 1 \ 7 \ 2 \ 5 \\ \hline 2 \end{array}$$

An algorithm (aka. program) from high school:

Addition of Large Numbers

$$\begin{array}{r} 73407 \\ + 61725 \\ \hline 2 \end{array}$$

The diagram illustrates the addition of two large numbers, 73407 and 61725, with a carry of 1. The numbers are aligned by their rightmost digits. A horizontal line is drawn under the second number. The result, 2, is shown below the line. A vertical orange box highlights the column containing the carry (1) and the digits 0 and 2, indicating the current step in the addition process.

An algorithm (aka. program) from high school: Addition of Large Numbers

73407
+ 61725

32

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} \\ \\ + \\ \hline \end{array}$$

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 73407 \\ + 61725 \\ \hline 32 \end{array}$$

The diagram illustrates the addition of two large numbers, 73407 and 61725, using a high school algorithm. The numbers are aligned by their rightmost digits. A vertical orange box highlights the third column from the right, which contains the digits 4 and 7. Above the 4 is a red '0', and above the 7 is a red '1', representing the carry-over from the previous step. The result of the addition, 32, is shown below the horizontal line, aligned under the third column.

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 1 1 \\ 7 3 4 0 7 \\ + 6 1 7 2 5 \\ \hline 1 3 2 \end{array}$$

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 101 \\ 73407 \\ + 61725 \\ \hline 132 \end{array}$$

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} \\ \\ + \\ \hline \end{array}$$

Diagram illustrating the addition of two large numbers (73407 + 61725) using a carry array. The carry values are shown in red above the digits:

Carry	7	3	4	0	7
1	0	1			

The sum of the two numbers is 132.

An algorithm (aka. program) from high school: Addition of Large Numbers

A vertical addition problem. The first number is 0101 (with leading zeros), the second number is 73407, and the result is 61725. The second column (1+3) is highlighted in orange.

	0	1	0	1	
	7	3	4	0	7
+	6	1	7	2	5
		5	1	3	2

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} \\ 7 3 4 7 \\ + 6 1 7 2 5 \\ \hline 5 1 3 2 \end{array}$$

An algorithm (aka. program) from high school: Addition of Large Numbers

A vertical addition problem. The numbers are aligned to the right. The first column (ones) is highlighted in orange. The numbers are 076 and 13407. The sum is 5132.

	0	1	0	1	
	7	3	4	0	7
+	6	1	7	2	5
		5	1	3	2

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 10101 \\ + 73407 \\ + 61725 \\ \hline 35132 \end{array}$$

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An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 0 0 \\ + 6 7 5 \\ \hline 1 5 3 \end{array}$$

The diagram illustrates the addition of two large numbers. The first number is 10101, and the second number is 73407. The sum is 135132. The carry values are shown in red above the digits of the first number. The plus sign is to the left of the second number. A horizontal line separates the addends from the sum.

An algorithm (aka. program) from high school: Addition of Large Numbers

$$\begin{array}{r} 10101 \\ 73407 \\ + 61725 \\ \hline 135132 \end{array}$$

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Building block:
Addition of single-digit
numbers

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Building block:

Addition of single-digit numbers

Repeat until:

All digits are exhausted

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$$\begin{array}{r} 10101 \\ 73407 \\ + 61725 \\ \hline 135132 \end{array}$$

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Addition of single-digit numbers

Repeat until:

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The steps (algorithm) is **independent** of the **input** as well as **input size**!

Some Basic Operations in a program

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Get data from keyboard, file, etc.

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Display data on screen, save in file, etc.

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Perform basic math operations. **Eg.** +, -, *, / , min, max.

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- **Conditional execution**

Check for certain conditions. **Eg:** Is $35 < (8 * 4)$.

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- **Repetition**

Perform some action repeatedly, usually with some variation.

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1. Sequence of simple but precise **instructions**.
2. **Flow of control**: what is the next step to be executed.
3. **Stopping criteria**: specifies when to stop.

First Informal Program:

Largest Power of 2 less than $n=100$

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Power

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** $\leq$ 100 ?

1

First Informal Program:

Largest Power of 2 less than $n=100$

Power

1

Is **Power** $\leq$ 100 ?



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64



128

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64



128



First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64



128



Answer: 64

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64



128



Math operations ?

Answer: 64

First Informal Program:

Largest Power of 2 less than $n=100$

Power

Is **Power** ≤ 100 ?

1



2



4



8



16



32



64



128



Math operations ?

Repeat until _____ ?

Answer: 64

Second Informal Program:

GCD of two numbers - 8 and 12

Second Informal Program:

GCD of two numbers - 8 and 12

num

Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



6



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



6



7



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



6



7



8



Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



6



7



8



Answer: 4

Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



6



7



8



Math operations ?

Answer: 4

Second Informal Program:

GCD of two numbers - 8 and 12

num

Does num **divide** 8, 12?

1



2



3



4



5



6



7



8



Math operations ?

Check numbers up to _____ ?

Answer: 4

Lots of puzzles, but first ...

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Need to learn

- how to read inputs,
- how to write outputs,
- what basic operations are available,
- how to store intermediate computations

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- how to read inputs,
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... will discuss this in the next lecture.